EVIDENCE COMPANION · EDITION 2.0.0

AGI ALPHA FLYWHEEL — When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

Vincent Boucher — President of MONTREAL.AI

1. Read the claims at their actual evidence level

This companion is a map from the paper's conclusions to executable artifacts. Paths below are relative to `studies/agi-alpha-evidence-construction-v2/`. The paper and earlier editions remain separate. No research acceptance is converted into a customer signature, invoice, collection or production authority.

Claim	Evidence label	Artifact and limit
Published table conflicts	Observed source + computed check	Publisher XML/PDF and original S 4 document; main marginal exceeds both subgroup means. This does not identify the true correction.
Conditional bounds	Executed computation	`results/development/constructed-delivery.json`; independently checked exact-rational means. No causal or sampling interval claim.
Generic instrument works in its scope	Executed, bounded confirmation	`validation/independent-confirmation/`: 32/32 checks. Does not automate semantic truth.
Fresh recipient transfer	Executed artifact use	`assay/recipient_eplus/` and replay tracing. Two unfamiliar reports; 20/20 for each U/E/E+. No complete-cost advantage measured.
Commercial feasibility	Assumed and simulated	`economics/`: buyer values, effort, prices, quality probabilities, collections and profit. No real customer trial.
Protected transfer/accounting	Existing code, fixture identities	`integration/results/`: actual bytes and local signatures; modeled evidence and identities. No live ENS or settlement.
Market/model context	Sourced or estimated	Official model documentation and primary prior art; same-scope human hours/rates explicitly estimated.

What is connected, and what is not silently equated

One actual development packet discovers and diagnoses the source problem. It motivates a frozen generic capsule. A separate recipient uses those exact recovered bytes on two fresh reports. A commercial model binds the packet type, delivery acceptance obligations and retained artifact to three initial and two later hypothetical orders. Those five model contracts are *not* five observed paid deliveries. The linkage receipt records the exact files and hashes for every step.

Actual research construction is unpaid C work with unknown complete cost. The commercial scenario's USD 125 A construction is a separate hypothetical cost of a paid outcome. Complete actual profit remains null. Neither the modeled USD 568.35 nor the USD 23.30 increment is earned revenue.

Decision supported now

Keep the research artifact and its limits. The actual trial supports executable transfer and correct bounded work, with no observed quality increment. A conditional service is economically testable. Monetary admission should wait for metered incremental savings and credible reuse or demand; ordinary expert delivery can proceed if its own scope, quality, cost and buyer conditions are satisfied.

2. The transformation witness and its primary sources

Original purpose. Produce a faithful, source-qualified descriptive evidence packet from a named report. An exact corrected-point request retains its original acceptance requirement.

Binding restriction. Printed aggregate means, subgroup means and participant counts cannot all describe the same stated population under their ordinary interpretation.

Removable assumption. Direct reuse need not mean selecting a printed aggregate as authoritative. Source hypotheses can be represented separately and tested, with an explicit strict hold when all supplied claims cannot coexist.

Replacement and execution. Enumerate positive integer cell counts, constrain latent means by displayed rounding, and bound the desired descriptive contrast. Preserve every omitted claim in each conditional branch. Give the same formulation to a conventional exact-rational verifier.

Falsifier. A silent source repair, hidden exclusion, wrong population, missing covariance promoted into an interval, or failure of the independent arithmetic would defeat the supported claim.

Development source and observations

Ritchie, Della Sala and McIntosh (2013), *Retrieval Practice, with or without Mind Mapping, Boosts Fact Learning in Primary School Children*. doi:10.1371/journal.pone.0078976. Table 2 is on PDF page 5. The original supplementary S 4 is preserved and rendered for independent visual review.

At five weeks, subgroup means 52.82 and 53.30 cannot produce marginal 54.07. Half-unit rounding at two decimals leaves a minimum inconsistency of 0.760 percentage points. S 4 counts (49, 50, 46, 41) produce column totals 95/91, whereas the main table reports 100/86. Both have row totals 99/87. Mean-consistent main-table counts (52, 47, 48, 39) also expose an SD inconsistency: a 23.578 lower bound exceeds the printed SD's maximum rounding-compatible value, 23.045.

Interpretation	Accepted status
All five-week main means, including 54.07	Inconsistent hold; no corrected point admitted.
Main-table mean branch	7.6485684431–7.6678787879 percentage points. Excludes the conflicting marginal; does not validate S 4 counts or SDs.
S 4 counts + cell means	7.5005851620–7.5205851620 percentage points. Does not adopt conflicting main-table marginals.

Both intervals describe rounding uncertainty only. No author-endorsed erratum, adjusted ANCOVA result, causal effect or confidence interval is claimed. The first pilot's unqualified SE fields remain development outputs. No source author was contacted. A targeted search cannot prove that no correction or earlier discovery exists.

Source identity anchors

XML SHA256: 18acae39e5cfbb8832a6ab616f49de795af8130ef9b1be52e03028febb202cb2.

PDF SHA256: 569342fd49d85f6d5b17b9a4752b80bba98f53f93e9c7f1ae5197472305dbf44.

S 4 DOC SHA256: 5f3497749584fb0ad203d84e9d720f7917f0945427e104d6f3a43fbc13df51e6.

3. Freezes, fresh work and contamination boundaries

The selected opportunity preceded the generic engine. The engine and expert method were frozen at 13:26:23 Toronto time, 27 October 2026 (17:26:23 UTC). A separate evaluator then froze the confirmation plan and seed before creating cases. A fresh curator selected the two primary reports without seeing the development answers or engine and froze its answer key before recipient sessions.

Freeze or check	Contents and result
Construction	`protocol/construction-freeze-v1.json`. No post-confirmation edits to the retained engine or method.
Independent confirmation	Seed 791430271: 12 fresh numeric cases, 4 contradictions, 4 pooled-moment comparisons, 8 numeric/scope cases, 4 source-boundary cases. 32/32 pass.
Recipient conditions	`protocol/recipient-confirmation-freeze.json`. Same inherited model/effort; at most 10 underlying tool operations and 12 minutes for both reports.
Task A	Maraver et al. (2022), doi:10.1371/journal.pone.0272427. Experiment 1 final correct responses: retrieval with feedback minus restudy with feedback.
Task B	Sun et al. (2015), doi:10.1371/journal.pone.0143177. Long-term minus short-term standardized peer-assessment effect; uncertainty must respect dependence.

Actual recipient results

Condition	A	B	Total	Calls	Recorded elapsed
U: ordinary	10/10	10/10	20/20	10	175.698 seconds
E: expert method	10/10	10/10	20/20	10	236.061 seconds
E+: method + capsule	10/10	10/10	20/20	9	235.150 seconds

All recover Task A's 0.22 proportion-unit difference (22 percentage points) and approximate Welch 95% interval [0.1343, 0.3057] under the qualified independent-participant interpretation. They distinguish the counterbalanced constructed restudy sequence from participant yoking. For Task B, all obtain 0.122 minus 0.115 = 0.007 and withhold an interval without joint estimator covariance; they distinguish participant, compliance and outcome-analysis sample sizes.

Unchanged script replays exactly match saved numerical outputs. E+ actually invokes `read_jats` twice, `cell_value` four times and `mean_difference` twice. It uses source reading and qualified arithmetic, not the development report's specialized four-cell bound function.

What these checks cannot establish

Each configuration has one session covering the same two papers; there is no population effect estimate. Worklog timing is internally checked, not independent provider telemetry or billed service effort. Exact token usage and actual complete costs are unavailable. Model pretraining may contain these public papers. Fresh conversational sessions and explicit file boundaries provide procedural isolation, not a separate security domain. The evaluator is independent of construction/recipient execution but knows condition labels and is an agent, not a blinded human panel.

The answer key was released after completion for reproducibility and remains outside the generic capsule. The archive contains no customer secret, primary article or evaluation key. Negative/zero gains remain in commercial sensitivity rather than being ruled out by a polished prompt.

4. Complete economics and the prospective correction

All dollars below are modeled USD. The reference quote is 325, selected from a frozen grid. The same-scope human service is estimated at 150 minutes and 125/hour, plus tools/support, for 332 total. U/E/E+ delivery efforts are assumed 110/90/70 minutes. These are not inferred from the short recipient sessions or imposed as necessary human-review time.

Cost class	Included once
A	Intake, extraction, verification, correction, packaging, model, tools, compute, storage, network, overhead and support. Necessary generic construction is 125 in the first paid cohort.
B	Additional assessment 30, rights/packaging/transfer 15 and upkeep 15: total 60. No repeat charge for A construction.
C	Additional bounded discovery 25. An optional 100-dollar Gym campaign is rejected in the reference; no mandatory training spend.

B/C are assumed outsourced fees. The internal 900-minute capacity applies to A; supplier capacity is not tested. Customer effort is a separate participation constraint (at most 15 minutes), not another cash expense. Labor is not charged twice as an opportunity cost.

Configuration	Accepted	Collections	A	B	C	Profit
U	3	975.00	757.98	0	25	192.02
E	5	1,625.00	1,054.95	0	25	545.05
E+ / same library	5	1,625.00	971.65	60	25	568.35

Every buyer and collection is simulated. Initial capital is 1,000; reserve is 100. Refundable 50-dollar deposits do not finance construction. E+ requires 757.99 opening cash. Its first three orders contribute 317.01 after A/C; E contributes 317.03. Paying B leaves E+ 257.01, enough for the next 169.33-dollar packet. Final E+ cash is 1,568.35 with 100 reserved and zero reference obligations. The optional 20-dollar group-coordination fee occurs only in the separate factorial, not these independently placed orders.

Corrections remain visible

V1 charges extra failed-correction and support labor but misses its minutes in the capacity counter. V1.1 was separately frozen, then tested on eight new capacity boundary cases: 8/8 pass. When preacceptance extra work cannot fit, the affected contract is cancelled, the deposit refunded and sunk work preserved. All 17 reference cash/acceptance results remain identical at 900 minutes. Common-failure, partial-delivery and support totals become 333, 303 and 378 minutes, replacing 288. Successful E+ remains 440 minutes.

The price screen originally used $p_a p_c q$, omitting retained accepted deposits under default. The exact expression $p_a[d + p_c(q - d)]$ still selects 325, increasing expected contribution by 0.95 per order. The event ledger was already correct; no realized profit is recovered. Earlier code, 72 worlds and failures remain unchanged. The fresh correction checks do not validate real demand.

Failure and investment thresholds

The 72 fresh assumed worlds contain 3 E+ wins, 57 ties and 12 losses relative to E, from -415.54 to $+438.00$. Common failure loses 751.74; late collection blocks subsequent work. B needs more than 14.4 minutes saved per use for two uses, or 7.2 for four credible uses. Core+B over five needs 17.76. The 60-total-minute urgent contract needs at least 36 minutes saved relative to E. No threshold is a measured benefit.

5. Capability identity, inherited system and rights

The generic capsule is an Apache-2.0 research component with four files: `evidence_engine.py`, `expert_method.txt`, `README.md` and `manifest.json`. It uses standard XML parsing, pooled statistics, integer enumeration and SciPy HiGHS. It is a constructed workflow/instrument; no claim of algorithmic novelty, proprietary-weight training or specialist ASI is made.

Exact retained identity

Archive: 6,674 bytes, SHA256

32a9c193200f43c94b922c9b7f5460a60a57f44031f1b2c88c8a50b15766a930.

Engine SHA256

c8e40133ab0269e67afb1eb40aecd3952864b13370de0e548b638dc6726c60cb.

Expert-method SHA256

53844baeaeef75983b9182cebde70fc802dc6df9a74ae73392fb51b28dd23c4f.

The recovered archive has identical bytes. A valid generic computation does not establish correct input meaning. Missions must recheck endpoint, population, timepoint, units, dependence, source identity and acceptance. Production authority is separately granted; current success cannot revive revoked permission.

Existing interfaces, bounded execution

The study verifies and reuses the predecessor's 29-file source closure. `createModeledNetwork` admits and transfers under explicitly modeled evidence. Unknown membership, no gain, negative transfer and an available equivalent reject the relevant path; stronger Beta invalidates reuse. AES-GCM protects the actual archive in the fixture delivery. Local Ed25519 Journal/TrustRegistry/CostLedger execution reconciles the modeled totals. Unknown actual costs block a complete profit seal, including the preserved initial refused attempt.

No live ENS identity, Jobs validator decision, governance action, customer permission or settlement was executed. The production scope remains unchanged. The capsule is not relabeled as `bounded-public-source-selection/v1` to force admission. An ordinary library with identical content and permissions ties the model; graph-specific monetary benefit is not claimed. Revocation stops mediated access, not deletion of a copy already received.

Source and version ledger

Prior study parent: 9653febcf751b4fc5a47bef1c1c8494d22f3dc75.

Earlier corrected research: 231834269599d85e49cfe1ca9917a1a874522d44.

Runtime adapter lineage: 80b345828f34fd8a6e592977dd17ef37141305f4.

Main/publication policy inspected: 47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4.

The original foundational economy's successful contribution (300,000), expected contribution (120,000) and required opening cash (900,000) remain different synthetic quantities. They are not market evidence. The original vision motivates discovery without validating market-size, IP-value or token claims.

Publication rights and disclosure

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Research, software, independent agent checks, drafting and French translation were AI-assisted. The requested attribution does not assert manual author execution, human peer review or actual commercial endorsement. New HTML remains subject to the repository's full-source publication qualification.

6. Reproduce, inspect and decide what comes next

Package map

Directory	What to inspect
`discovery/`	Three-candidate dossier, current-model documentation, source and market uncertainty.
`protocol/`	Selection, transformation witnesses, lineage, construction and recipient freezes.
`data/development/`	Exact publisher bytes, fetch receipts and named source hypotheses.
`instrument/`	Frozen executable component and capsule identity.
`assay/`	Full ordinary/expert prompts, public source packet, original answers, calculation scripts and worklogs.
`validation/`	Exact verifier, fresh confirmation, adjudication, linkage and visual/browser checks.
`economics/`	Frozen assumptions, history, v1 results, v1.1 capacity correction, cash paths and residual-obligation sensitivity.
`integration/`	Pinned component verification, actual capsule transfer, native modeled accounting and unknown-actual-cost refusal.
`paper/`, `web/`	LaTeX/PDF, brief, this companion, full browser source and interactive EN/FR demonstration.

Run the provided portable reproduction entry point in an isolated output directory and preserve the original outputs. The Python calculations require NumPy/SciPy; native integration requires Node and the pinned predecessor closure. PDF builds require XeLaTeX/BibTeX with DejaVu fonts, and the browser builder uses Pandoc. Source documents are already bundled where rights are verified; network access is not needed for their core calculations. Optional unpublished raw companions are not required to recompute the accepted printed-summary quantities.

Original recipient scripts contain historical paths and remain unchanged. Portable replay uses explicit temporary path mappings, recorded in its receipt. Replaying calculations is not rerunning the model conversations. Reproducing the 72 worlds demonstrates the declared simulation, not its external calibration. Scoped checks are reported honestly; a reduced research snapshot is not a fully qualified production repository.

Preserved failures and negative evidence

The strict source branch remains infeasible. The initial development SE outputs are not admitted as adjusted inference. The fresh U/E/E+ quality comparison ties. Most economic worlds tie or lose; graph and library tie. Optional Gym and no-gain/available-equivalent retention can be rejected. An initial native complete-cost seal refuses unknown costs. A one-cent residual-analysis rounding error and the capacity/pricing issues have separate correction records. The first narrow-screen demo overflow was repaired and retested. These records prevent a successful final package from erasing its learning history.

The smallest next experiment

Recruit a qualified analyst or customer only with an agreed, reviewable narrow scope. Obtain an equivalent human-service quote. Randomize E/E+ order across unfamiliar reports; keep the accepted quality fixed. Meter provider costs, active effort, retries, source clarification, support, customer effort and completion time. Count failures as work and keep the selected artifact frozen. The key decision is whether complete savings exceed 14.4 minutes for two fresh uses, or another threshold justified by a credible reuse forecast and actual costs.

If the service is profitable but retention is not, continue expert delivery and stop the extra investment. If an already available tool ties it, use that tool. If greater model defaults absorb the saving, requalify the method as ordinary Beta. Build another instrument only when a new consequential gap and bounded experiment justify it.

The supported achievement is precise: a real source problem became an auditable conditional answer, its constructed instrument transferred to fresh work, and a complete economic model exposes the conditions under which that inheritance would pay. The unmeasured increment remains the next question, not an earned surplus.